

Dr. Sergio Ruocco

Senior Software Engineer, EMASS · Embedded and real-time systems, compilers
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Research profile

Dr. Ruocco's research concerns the analysis, design and implementation of time-sensitive and time-aware computer systems. His PhD introduced temporal reflection, a class of computational reflection through which a system can represent, observe and control its own temporal behaviour; he designed and implemented a reflective real-time scheduler for the L4 microkernel. At NICTA he analysed and improved the real-time properties of general-purpose L4 microkernels. At A*STAR and NTU he worked on systems software and programming toolchains for non-volatile memory, storage systems, RISC-V platforms and AI accelerators, including the port of MbedOS and Linux device drivers to the ECS-DoT SoC and its AI accelerator. He is currently a Senior Software Engineer at EMASS, building the full software stack (from the development boards' boot-agent firmware to the SDK), the CI/CD build and test pipeline, and the IT/DevOps infrastructure for the ECS-DoT ultra-low-power AI SoC used by the EMASS engineering team and its customers; he planned its Zephyr RTOS port and delivered it with a newly formed team he mentored. His work has appeared at IEEE RTSS, ISCAS, TCAS-I and JETCAS and in the EURASIP Journal on Embedded Systems, and has led to a US patent and several technology disclosures licensed to EMASS Pte Ltd.

Research interests

- Real-time and embedded systems
- Temporal reflection and reflective scheduling
- Operating-system kernels and microkernels (L4)
- Linux kernel and device drivers
- Compilers and programming toolchains (GCC)
- Computer architecture
- Non-volatile memory
- Storage systems and performance analysis
- Energy-aware computing
- RISC-V platforms and AI accelerators
- Software stacks for ultra-low-power AI SoCs (ECS-DoT)

Education

2004 — PhD in Computer Science
Università degli Studi di Milano, Italy — Mathematics, Statistics, Computational Sciences and Computer Science Research

Thesis: *Temporal Reflection*. Doctoral scholarship, 2000–2004. In Italy, PhD degrees are pass/no-pass and are not graded.

1999 — Master of Science in Computer Science
Università degli Studi di Milano, Italy — 110/110 cum laude

Thesis: *HyperReal: a paradigm for the design and synthesis of hard real-time control systems*.

Experience

Senior Software Engineer — EMASS (an NTU/A*STAR spin-off)
06/2025 – present · Singapore

End-to-end software for ECS-DoT, an ultra-low-power AI SoC with a RISC-V core and dedicated hardware accelerators. Research and development of the full stack of software and tools — device drivers, RTOS, SDK and toolchains — to compile, optimise and run AI applications on a custom SoC designed and fabricated by the team.

- Planned the port of the Zephyr RTOS to ECS-DoT and, once approved, delivered it on time by training and mentoring a newly formed small team; packages the Zephyr SDK (board support, containers, debug tooling).
- Develops and releases the bare-metal SDK and runtime of the ECS-DoT ultra-low-power AI SoC; integrates and qualifies the GCC RISC-V cross-toolchains (GCC 7 and GCC 11, x86-64 and arm64 hosts).
- Targets a RISC-V core with dedicated deep-learning accelerators for edge AI IoT applications.
- Built the ECS-DoT boot agent on the open-source DirtyJTAG firmware for the Raspberry Pi Pico 2 (RP2350) of the ECS-DoT development boards: it loads RISC-V applications into the SoC over JTAG at power-up, doubles as an

OpenOCD adapter and USB-serial bridge, controls board power, and has shipped to customers in tagged releases since 07/2026.
- Fixed and extended the OpenOCD DirtyJTAG adapter driver (protocol, bit ordering, multi-adapter selection, reset lines) to drive the SoC's RISC-V debug module; designed the boot agent's companion link to the ECS-DoT runtime for dynamic voltage and frequency scaling and on-board power measurement of AI inference.
- Designed and runs the SDK build-and-release pipeline: a three-tier container hierarchy (Linux devtools, vanilla bare-metal and Zephyr SDKs, per-customer SDK images) on self-hosted GitLab CI with cross-project rebuild cascades, published to GitHub and GitLab registries; regular bare-metal SDK releases since v4.0.0 (04/2026).

Research Fellow — School of Computer Science and Engineering, Nanyang Technological University (NTU)

04/2024 – 05/2025 · Singapore

Kept the ECS-DoT system-software engineering going between A*STAR I²R and EMASS, laying the groundwork for the EMASS role: the first bare-metal SDK, its SPI and camera drivers, and analysis of hardware bugs in the SoC's SPI peripheral.

- Developed the first bare-metal SDK for the ECS-DoT SoC.
- Completed the bit-bang SPI driver and ported an SPI camera driver to the bare-metal SDK.
- Isolated and analysed hardware bugs in the SoC's hardware SPI peripheral.

Senior Scientist II — A*STAR Institute for Infocomm Research (I²R)

06/2018 – 09/2023 · Singapore

Embedded real-time operating systems, SDKs, firmware and drivers for a RISC-V-based SoC customised with a deep-learning accelerator for ultra-low-power edge-IoT applications.

- Delivered the port of ARM MbedOS 5.12.4 and its SDK to the ECS-DoT RISC-V SoC, including CPU core, peripherals, memory map and I/O devices.
- Ported the AI accelerator's ARM Linux kernel driver to MbedOS on the ECS-DoT RISC-V CPU, and traced the Linux compiler and kernel driver for the accelerator.
- Containerised the ECS-DoT SDK build with Docker, including OpenOCD and JTAG/USB board access (2023): the starting point of the SDK build-and-release pipeline later run at EMASS.
- Contributed to the Edge SoC with AI acceleration technology disclosure (TD2022128) and three software disclosures, all licensed to EMASS Pte Ltd.

Scientist II — A*STAR Data Storage Institute (DSI)

10/2012 – 05/2018 · Singapore

Systems software and programming-toolchain research for next-generation solid-state storage and non-volatile memory.

- Extended Linux memory management to support non-volatile DIMMs; the work led to an invited collaboration with CERN OpenLab and a talk at CERN OpenLab 2015.
- Led and delivered the Active Discs Clusters Performance Study (ADCPS) for Seagate USA (2017–2019), completed after the move to I²R.

Research Fellow — Dipartimento di Informatica, Sistemistica e Comunicazione (DISCo), Università di Milano-Bicocca

04/2007 – 09/2012 · Milan, Italy

Research and teaching in real-time embedded systems, compiler retargeting and toolchains, and next-generation non-volatile storage, with projects funded by industry contracts and European research grants.

- Principal Investigator and technical lead for UNIMIB in EU ENIAC SMART (€302,400) and TOISE (€556,000), with Micron Italy and STMicroelectronics.
- Secured two Micron R&D contracts, including the port of GCC to a custom 16-bit microcontroller for phase-change memory chips.
- Project leader of IMUD/Cyber-ADI, a long-running, low-power wireless real-time inertial monitoring device for wireless sensor-network research.
- Taught graduate Computer Architecture and undergraduate Real-time Embedded Systems.

Researcher, Embedded, Real-Time and Operating Systems (ERTOS) Programme — NICTA (National ICT Australia, now Data61/CSIRO), Sydney

04/2004 – 03/2007 · Australia

Real-time and operating-systems research on the L4 microkernel.

- Analysed and improved the real-time properties of the general-purpose L4 microkernel.
- Designed and implemented a reflective real-time scheduling framework (RTSS 2006; OSPERT 2006/2007; EURASIP JES 2008).
- Visiting Fellow and conjoint lecturer at the University of New South Wales (2004–2007).

Grants and funded projects

ENIAC TOISE — Trusted Computing for European Embedded Systems

01/2011 – 12/2013 · €556,000 · Principal Investigator and technical lead for UNIMIB

Partners: Micron Italy, STMicroelectronics and European consortium partners.

EU ENIAC Joint Undertaking project on trusted computing for embedded systems, including non-volatile-memory security.

Project page

ENIAC SMART — Techniques and Tools for Advanced Real-Time Embedded Firmware

01/2010 – 12/2012 · €302,400 · Principal Investigator and technical lead for UNIMIB

Partners: Micron Italy, STMicroelectronics and European consortium partners.

EU ENIAC Joint Undertaking project on techniques and tools for advanced real-time embedded firmware design and development.

Project page

Micron R&D contracts

2009 – 2014 · Principal investigator and lead engineer

Partners: Micron Italy (formerly STMicroelectronics).

Two consulting research contracts: a GCC port to a custom 16-bit microcontroller for phase-change memory (2009–2010), and software architectures for non-volatile storage (2012–2014).

Active Discs Clusters Performance Study (ADCPS)

01/2017 – 05/2019 · Project lead

Partners: Seagate USA.

Industry-funded performance study of active disc clusters.

CERN OpenLab collaboration

2015 · Invited collaborator

Partners: CERN OpenLab.

Collaboration on non-volatile memory and persistent data in main memory, following the Linux memory-management work at A*STAR DSI.

Project page

Selected projects

Full software stack for the ECS-DoT ultra-low-power RISC-V AI SoC

*2024 – present · NTU, then EMASS (an NTU/A*STAR spin-off) from 06/2025*

Research and develop the device drivers, RTOS, SDK and toolchains needed to compile, optimise and run AI applications on ECS-DoT, a custom ultra-low-power SoC with a RISC-V core and dedicated hardware accelerators, designed and fabricated by the team.

Impact: Enables AI workloads on the ECS-DoT SoC; builds on the AI-accelerator driver and RTOS work done at A*STAR I²R.

ECS-DoT boot agent and JTAG adapter on the Raspberry Pi Pico 2

2026 – present · EMASS · ECS-DoT development boards

Extended the open-source DirtyJTAG firmware into the boards' boot agent: it loads RISC-V applications into the SoC over JTAG at power-up, serves as an OpenOCD adapter and USB-serial bridge, and drives the boards' power control and measurement.

Impact: Shipped to customers in tagged firmware releases since 07/2026.

MbedOS and SDK port to the ECS-DoT RISC-V SoC

*2020 – 2024 · A*STAR I²R / NTU · ECS-DoT Edge SoC with AI acceleration*

Ported the ARM MbedOS 5.12.4 RTOS and its SDK to the ECS-DoT SoC, covering a different CPU core, instruction set, peripherals, memory map and I/O devices. Both the original and the target platforms were poorly documented and buggy, so the work was a complex trial-and-error effort with the GCC toolchain and GDB.

Impact: Delivered a working RTOS port and board-support package; technology disclosure SWIP-PMEP-2022-31, licensed to EMASS Pte Ltd.

AI accelerator Linux kernel driver port to MbedOS on RISC-V

*2022 – 2023 · A*STAR I²R · ECS-DoT*

Ported the AI accelerator's ARM Linux kernel driver to MbedOS on the ECS-DoT RISC-V CPU, adapting the driver to a new operating system, memory map and device model.

Impact: Technology disclosure SWIP-PNVD-2022-33, licensed to EMASS Pte Ltd.

Real-time extensions and reflective scheduling for the L4 microkernel

2004 – 2007 · NICTA · ERTOS Programme

Analysed the real-time behaviour of general-purpose L4 microkernels, identified the architectural obstacles to predictable scheduling, and designed and implemented a reflective real-time scheduler that lets a system observe and control its own temporal behaviour.

Impact: Published at IEEE RTSS 2006 and OSPERT 2006/2007 and in the EURASIP Journal on Embedded Systems; the work influenced the real-time features of the commercial OKL4 microkernel.

Linux memory-management extensions for non-volatile DIMMs

*2015 · A*STAR DSI · CERN OpenLab collaboration*

Extended Linux memory management to support non-volatile DIMMs, allowing persistent data to be placed directly in main memory and processed at memory speed.

Impact: Procured an invitation to collaborate with CERN OpenLab and an invited talk at CERN OpenLab 2015.

GCC backend for a custom 16-bit microcontroller

2009 – 2010 · Micron Italy (formerly STMicroelectronics) · Università di Milano-Bicocca

Ported the GNU GCC C compiler to a custom, non-standard 16-bit microcontroller managing embedded NAND/NOR phase-change memories under hard real-time constraints and security requirements. The target was poorly documented, requiring a new backend and extensive debugging with GDB.

Impact: Delivered under a Micron R&D contract; the compiler supported devices deployed in high volume.

EU ENIAC SMART and TOISE projects

2009 – 2013 · Università di Milano-Bicocca · Micron Italy and STMicroelectronics

Principal Investigator and technical lead for UNIMIB in two three-year EU ENIAC projects: SMART (techniques and tools for advanced real-time embedded firmware, €302,400) and TOISE (trusted computing for European embedded systems, €556,000). Responsible for planning, budget, recruiting research engineers and post-docs, and scientific and technical delivery.

Impact: Funded a research fellowship and a team of post-docs and research engineers; produced tools and techniques for advanced embedded firmware.

Publications

5 journal articles, 12 conference papers and 1 PhD thesis.

Journal articles

1. Q. Trinh, S. Ruocco, M. Alioto, "Time-Based Sensing for Reference-Less and Robust Read in STT-MRAM Memories," *IEEE Transactions on Circuits and Systems I: Regular Papers*, 65(10):3338–3348. DOI · 39 citations
2. Q. Trinh, S. Ruocco, M. Alioto, "Dynamic Reference Voltage Sensing Scheme for Read Margin Improvement in STT-MRAMs," *IEEE Transactions on Circuits and Systems I: Regular Papers*, 65(4):1269–1278. DOI · 33 citations
3. Q. Trinh, S. Ruocco, M. Alioto, "Novel Boosted-Voltage Sensing Scheme for Variation-Resilient STT-MRAM Read," *IEEE Transactions on Circuits and Systems I: Regular Papers*, 63(10):1652–1660. DOI · A*STAR Research highlight · 31 citations
4. Q. Trinh, S. Ruocco, M. Alioto, "Voltage Scaled STT-MRAMs Towards Minimum-Energy Write Access," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, 6(3):305–318. DOI · 22 citations
5. S. Ruocco, "A Real-Time Programmer's Tour of General-Purpose L4 Microkernels," *EURASIP Journal on Embedded Systems*, 2008:1–14. DOI · 17 citations

Conference papers

1. Q. Trinh, S. Ruocco, M. Alioto, "Novel Time-Based Sensing Scheme for STT-MRAMs," *IEEE International Symposium on Circuits and Systems (ISCAS)*, Florence, Italy, 1–5. DOI · 1 citation
2. Q. Trinh, S. Ruocco, M. Alioto, "STT-MRAM Write Energy Minimization via Area Optimization under Dynamic Voltage Scaling," *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2791–2794. DOI
3. Q. Trinh, S. Ruocco, M. Alioto, "Boosted Sensing for Enhanced Read Stability in STT-MRAMs," *IEEE International Symposium on Circuits and Systems (ISCAS)*, 1238–1241. DOI · 11 citations
4. S. Ruocco, D.-K. Le, "Efficient Persistence of Financial Transactions in NVM-based Cloud Data Centers," *International Conference on Cloud Computing Research and Innovation (ICCCRI)*, 25–36. DOI
5. Q. Trinh, S. Ruocco, M. Alioto, "Modeling the Impact of Dynamic Voltage Scaling on 1T-1J STT-RAM Write Energy

and Performance," *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2313–2316. DOI · 4 citations
6. K. Elphinstone, D. Greenaway, S. Ruocco, "Lazy Queueing and Direct Process Switch — Merit or Myths?," *Workshop on Operating Systems Platforms for Embedded Real-Time Applications (OSPERT)*, Pisa, Italy. paper · 2 citations
7. S. Ruocco, "User-Level Fine-Grained Adaptive Real-Time Scheduling via Temporal Reflection," *27th IEEE Real-Time Systems Symposium (RTSS)*, Rio de Janeiro, Brazil, 246–256. DOI · 8 citations
8. S. Ruocco, "Real-time Programming and L4 Microkernels," *Workshop on Operating Systems Platforms for Embedded Real-Time Applications (OSPERT)*, Dresden, Germany. paper
9. K. Elphinstone, G. Heiser, R. Huuck, S. Petters, S. Ruocco, "L4cars," *Embedded Security in Cars (ESCAR) Workshop*, Cologne, Germany. paper
10. D. C. Snowdon, S. Ruocco, G. Heiser, "Power Management and Dynamic Voltage Scaling: Myths and Facts," *Workshop on Power Aware Real-time Computing (PARC)*, New Jersey, USA. paper · 115 citations
11. F. Engel, G. Heiser, I. Kuz, S. M. Petters, S. Ruocco, "Operating Systems on SoCs: A Good Idea?," *Workshop on Embedded Real-Time Systems Implementation (ERTSI)*, Lisbon, Portugal. paper · 13 citations
12. D. Micucci, S. Ruocco, F. Tisato, A. Trentini, "Time Sensitive Architectures: A Reflective Approach," *7th IEEE International Symposium on Object-Oriented Real-Time Distributed Computing (ISORC)*, Vienna, Austria, 189–196. DOI · 7 citations

Thesis

1. S. Ruocco, "Temporal Reflection," *PhD thesis, Università degli Studi di Milano*, Milan, Italy.

Citation counts: OpenAlex (November 2026). Live counts and h-index: Google Scholar.

Patents and technology disclosures

US 16/234,876 — "Dynamic Reference Scheme for Improving Read Margin of Resistive Memory Array." Filed 2018-12-28; granted 2019-05-09. US20190206469A1 · Managed by the NUS IP office.

Technology disclosures

*EMASS (Embedded A.I. Systems Pte Ltd) is a Singaporean fabless semiconductor company founded in 2020 as a spin-off from NTU Singapore and A*STAR research, and acquired by Nanoveu Ltd (ASX: NVU) in March 2025.*

SWIP-PMEP-2022-31 — Porting of MbedOS 5.12.4 to the ECS-DoT SoC and RISC-V CPU. Filed 2023-03-10. Review completed 27 April 2023. Licensed to EMASS Pte Ltd.

SWIP-PNVD-2022-33 — Porting of the AI accelerator ARM Linux kernel driver for MbedOS on the ECS-DoT RISC-V CPU. Filed 2023-03-10. Review completed 2 May 2023. Licensed to EMASS Pte Ltd.

SWIP-TLCN-2022-32 — Tracing Linux compiler and kernel driver for an AI accelerator. Filed 2023-03-10. Review completed 21 April 2023. Licensed to EMASS Pte Ltd.

TD2022128 / I2R/Z/13931 — Edge SoC with AI acceleration. Filed 2023-01-12. Licensed to EMASS Pte Ltd.

Invited talks

- **June 2015** — "Speeding-up Large-Scale Storage with NVM," *CERN OpenLab 2015*, Geneva, Switzerland. event
- **July 2007** — "Temporal Reflection and Reflective Scheduling for the L4 Microkernel," *WIRTES Workshop on Real-Time Embedded Systems, ECRTS*, Pisa, Italy.
- **December 2006** — "Real-Time Programming and L4 Microkernels," *Electrical and Computer Engineering Department, Federal University of Rio de Janeiro*, Rio de Janeiro, Brazil.
- **July 2006** — "Time Sensitive Architectures: A Reflective Approach," *TU-Dresden*, Dresden, Germany.
- **July 2006** — "Real-Time Programming and L4 Microkernels," *Electronic Department, Politecnico di Milano*, Milan, Italy.

Teaching

- **2009 – 2012** — *Real-time Embedded Systems* (undergraduate), Università di Milano-Bicocca, DISCo. Four editions (2008/09, 2009/10, 2010/11, 2011/12).
- **2007 – 2011** — *Computer Architecture* (graduate), Università di Milano-Bicocca, DISCo. Multiple editions (2006/07 to 2010/11).
- **2008** — *Tecnologie Informatiche per la Comunicazione* (undergraduate), Università di Milano-Bicocca, DISCo. Operating-system technologies for mobile devices.
- **2005** — *COMP 3241/9245 — Real-time Systems: Design, Analysis, Implementation* (undergraduate and graduate), School of Computer Science and Engineering, University of New South Wales, Sydney. Uniformly high ratings in the anonymous student survey.

- **2000 – 2004** — *Computer Architecture and Networks Laboratory* (teaching assistant), Università di Milano-Bicocca, DISCo. Five editions.

Student supervision

- **PhD** — 1 SINGA PhD co-supervision with the National University of Singapore.
- **Internship** — 1 three-month research intern.

Professional service

- **2011** — Programme Committee member, *Brazilian Symposium on Computing Systems Engineering (SBESC)*, Florianópolis, Brazil. website

Technical skills

Operating systems and platforms. Linux (Debian, Ubuntu, Mint, Red Hat) on workstations and datacentre servers; L4 microkernels, MbedOS, Zephyr, RTOS internals and porting; Linux kernel, memory management and device drivers; ZFS and zfs-root workstations and storage servers; IPMI, RAID, GPU servers; Secure build systems and export-controlled software licence servers.

Languages and toolchains. C, GCC/G++, GDB and the GNU toolchain, native and cross-compiled; GCC backend and cross-compiler development; Buildroot, Android SDK, MbedOS SDK; Python, Bash, Git, SSH, GPG, tmux/screen; AI-assisted coding; Docker and Buildx, GitLab CI multi-project pipelines, GitHub and GitLab container registries; IT/DevOps infrastructure for engineering teams.

Architectures and hardware. ARM and RISC-V platforms; board bring-up, JTAG and RISC-V debug (OpenOCD); Microcontroller firmware on RP2040/RP2350: Pico SDK, PIO, TinyUSB, I²C and UART; FPGA boards, peripherals, memory maps and I/O devices; Non-volatile memory (NVM-DIMM, PCM, STT-MRAM) and storage systems; AI accelerators and video-compression hardware.

Methods. Real-time scheduling and temporal reflection; Performance analysis and system tracing; Debugging poorly documented hardware and software; Technical leadership, mentoring, procurement and project delivery.